

[Aim: 100/100 in Maths]

आरंभ

REAL NUMBERS

Lecture #5

#ATK²:

→ LCM & HCF

→ prime & composite

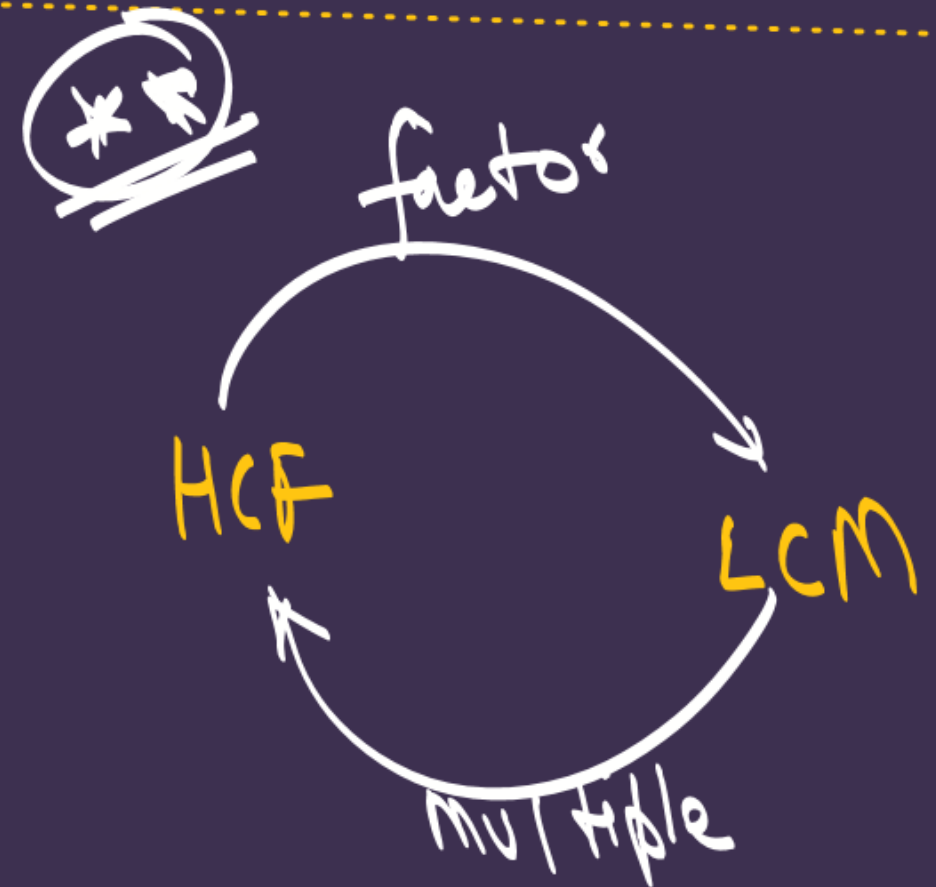
→ $(p \& q)$ → prime no. $\begin{cases} \text{HCF} = 1 \text{ (fix)} \\ \text{LCM} = p \times q \end{cases}$

→ FACTORS & MULTIPLE



↑ divide
9/1
3/2
9/3

↑ divide
9/1
3/2
9/3



#LP: Two numbers are in the ratio $2:3$ and their LCM is 180. What is the HCF of these numbers? [CBSE 2023]

let the numbers are $2k$ & $3k$

$$2k = 2 \times k$$

$$3k = 3 \times k$$

$$\boxed{\text{HCF} = k}$$

$$\text{LCM} = 2 \times 3 \times k$$

$$\boxed{\text{LCM} = 6k}$$

ATQ

$$\text{LCM} = 180$$
$$6k = 180$$

$$\boxed{k = 30}$$

final Ans $\rightarrow$

$$\boxed{\text{HCF} = k = 30}$$

Ans

#LP: Two numbers are in the ratio of 5:11. If their H.C.F. is 13, then numbers will be:

- (A) 195 and 143
- (B) 190 and 140
- (C) 185 and 163
- (D) 185 and 143

let the nos are $5k$ & $11k$

$$5k = 5 \times k$$

$$11k = 11 \times k$$

$$\boxed{HCF = k}$$

$$LCM = 5 \times 11 \times k$$

$$\boxed{LCM = 55k}$$

A7Q

$$HCF = 13$$

$$\boxed{k = 13}$$

$$\begin{aligned} \text{No.s} \rightarrow 5k &= 5 \times 13 = 65 \\ 11k &= 11 \times 13 = 143 \end{aligned}$$

the numbers are 65 & 143.

#LP: Find the smallest number which when increased by 11 is exactly divisible by both 198 and 486.

(A) 10681

✓ (B) 5335

(C) 5346 ✗

(D) 16027

$$198 = 2 \times 3 \times 3 \times 11 \Rightarrow 2^1 \times 3^2 \times 11^1$$

$$486 = 2^1 \times 3^5$$

$$\text{LCM} = 2^1 \times 3^5 \times 11^1$$

$$\boxed{\text{LCM} = 5346}$$

$$\textcircled{x} + 11 = 5346$$

$$x = 5346 - 11$$

Ans →

$$\boxed{x = 5335}$$

#LP: Find the smallest number which when ~~increased by 9~~ ^{decreased by 9} is exactly divisible by both 198 and 486.

(A) 10681

(B) 5335

(C) 5346

(D) 16027

$$198 = 2 \times 3 \times 3 \times 11 \Rightarrow 2^1 \times 3^2 \times 11$$

$$486 = 2^1 \times 3^5$$

$$\text{LCM} = 2^1 \times 3^5 \times 11$$

$$\boxed{\text{LCM} = 5346}$$

Smallest no. which is exactly div.

$$y - 9 = 5346$$

$$y = 5346 + 9$$

$$\boxed{y = 5355}$$

#LP: Find the smallest number which leaves remainders 18 when divided by 27 and 33.

(A) 297

(B) 279

✓ (C) 315

(D) 612

$$27 = 3^3$$

$$33 = 3 \times 11$$

$$\text{LCM} = 3^3 \times 11$$

$$\boxed{\text{LCM} = 297} \rightarrow \boxed{R=0}$$

ये दो नो. 297 पर divisible है।

Final Answer

$$= 297 + 18$$

$$= \boxed{315}$$

~~28~~

$$\begin{array}{r} 2 \\ 9 \sqrt{5} \\ \underline{45} \\ 1 \end{array}$$

$r = 0$
 $\leftarrow$
#

#LP: Determine the greatest 3-digit number exactly divisible by 8, 10, and 12. [CBSE 2017]

$$8 =$$

$$10 =$$

$$12 =$$

$$\text{LCM} = 120$$

$$120 \times 1 = 120$$

$$120 \times 2 = 240$$

$$120 \times 3 = 360$$

$$120 \times 4 = 480$$

$$120 \times 5 = 600$$

$$120 \times 6 = 720$$

$$120 \times 7 = 840$$

$$120 \times 8 = 960$$

$$120 \times 9 = 1080$$

largest 3 digit

Ans $\rightarrow$ 960

4 digit

#K³B:

for two numbers a & b :



$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

#LP: Given that $\text{HCF}(306, 657) = \underline{9}$, find LCM (306, 657).

$$\boxed{H \times L = a \times b} \checkmark$$

$$\cancel{9} \times L = \overset{34}{\cancel{306}} \times 657$$

$$L = 34 \times 657$$

$$\boxed{\text{LCM} = 22338}$$

#LP: Assertion (A): For any two prime numbers p and q , their HCF is 1 and LCM is $p + q$.

Reason (R): For any two natural numbers, $\text{HCF} \times \text{LCM} = \text{product of numbers}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

#LP: If the HCF (2520, 6600) = 40 and LCM (2520, 6600) = 252 x k, then the value of k is: [CBSE 2024]

- ✓ (A) 1650
- (B) 1600
- (C) 165
- (D) 1625

$$H \times L = 2520 \times 6600$$

$$40 \times \cancel{252} k = \overset{10}{\cancel{2520}} \times \overset{1650}{\cancel{6600}}$$

$$k = 1650$$

#LP: The HCF of two numbers 65 and 104 is 13. If LCM of 65 and 104 is $40x$, then the value of x is: **[CBSE 2024]**

- (A) 5**
- (B) 13**
- (C) 40**
- (D) 8**

#LP: The LCM of two numbers is 630 and their HCF is 9. If the sum of the numbers is 153, their difference is:

(A) 63

(B) 27 ✓

(C) 81

(D) 18

$$L = 630$$

$$H = 9$$

we know

$$H \times L = a \times b$$

$$9 \times 630 = a \times b$$

$$\boxed{5670 = ab}$$

$$a + b = 153$$

$$a - b = ??$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$-(a-b)^2 = -a^2 + b^2 - 2ab$$

$$(a+b)^2 - (a-b)^2 = 4ab$$

$$(a+b)^2 - 4ab = (a-b)^2$$

$$(153)^2 - 4(5670) = (a-b)^2$$

$$(a-b)^2 = \underline{729}$$

$$a-b = \sqrt{729}$$

$$\boxed{a-b = \pm 27}$$

Next DPP

THANK YOU COODIESS !!





Exercise-1

Part - 1 (Basic Multiple Choice Questions)

1. If prime factorization of 7560 is expressible as $2^3 \times 3^p \times q \times 7$, then the values of p and q are respectively ___ and ___
- (a) 2, 3 (b) 5, 3 (c) 3, 5 (d) 5, 2

5. The greatest number which when divides 1251, 9377 and 15628 leaves remainder 1, 2 and 3 respectively is
- (a) 575 (b) 450 (c) 750 (d) 625

$$\begin{array}{l} 1251 \longrightarrow R=1 \\ 9377 \longrightarrow R=2 \\ 15628 \longrightarrow R=3 \end{array}$$

HCF

Part - 2 (Basic Subjective Type Questions)

3. $P = 2(4)(6)\dots\dots\dots(20)$ and $Q = 1(3)(5)\dots\dots\dots(19)$. What is the HCF of P and Q.

$$P = 2 \times 4 \times 6 \times 8 \times 10 \times 12 \times 14 \times 16 \times 18 \times 20$$

$$Q = 1 \times 3 \times 5 \times 7 \times 9 \times 11 \times 13 \times 15 \times 17 \times 19$$

$$\text{HCF} =$$



Exercise-2

Part - 1 (Advanced Multiple Choice Questions)

6. Let a and b be two positive integers such that $a = p^3q^4$ and $b = p^2q^3$, where p and q are prime numbers. If $\text{HCF}(a, b) = p^m q^n$ and $\text{LCM}(a, b) = p^r q^s$, then $(m + n)(r + s) =$
- (a) 15 (b) 30 (c) 35 (d) 72

7. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then the result obtained by dividing the product of the positive integers by the LCM (a, b) is :

(a) xy

(b) xy^2

(c) x^3y^3

(d) x^2y^2

1. Prime factorisation of three numbers A, B and C is given below :

$$A = (2^r \times 3^p \times 5^q)$$

$$B = (2^p \times 3^r \times 5^q)$$

$$C = (2^p \times 3^q \times 5^r)$$

such that , $p < q < r$ and $p, q,$ and r are natural numbers.

- The largest number that divides A, B and C without leaving a remainder is 30.
- The smallest number that leaves a remainder of 2 when divided by each of A, B and C is 5402

Find A, B and C. Show your work.

Reattempt

4. P is the LCM of 2, 4, 6, 8, 10; Q is the LCM of 1, 3, 5, 7, 9 and L is the LCM of P and Q. Then, find the relation between L and P.

5. On the two real numbers $a = 2 + \sqrt{5}$ and $b = 3 - \sqrt{7}$, perform the following operations :

[CBSE CFPQ]

- (i) Calculate the sum $(a + b)$
- (ii) Calculate the product (ab)
- (iii) Find the additive inverse of a
- (iv) Rationalise $1/b$
- (v) Verify whether the numbers a and b are rational or irrational. Provide a valid reason for your answer

Part - 3 (Advanced Miscellaneous Questions)

6. Match the Column I with Column II and choose the correct option :

Column-I	Column-II
(A) The HCF of the smallest number of two digits and the largest multiple of 5 which is less than 40 is	(i) 5
(B) HCF of $x^3y^4z^2$ and $x^2y^3z^5$, where x, y and z are distinct prime numbers is	(ii) 4
(C) The total number of factors of a prime number is	(iii) 2
(D) The sum of exponents of prime factors in the prime factorisation of 196 is	(iv) $x^2y^3z^2$

- (a) A-iv, B-ii, C-i, D-iii
- (b) A-i, B-iv, C-iii, D-ii
- (c) A-ii, B-iv, C-i, D-iii
- (d) A-ii, B-iv, C-iii, D-i